

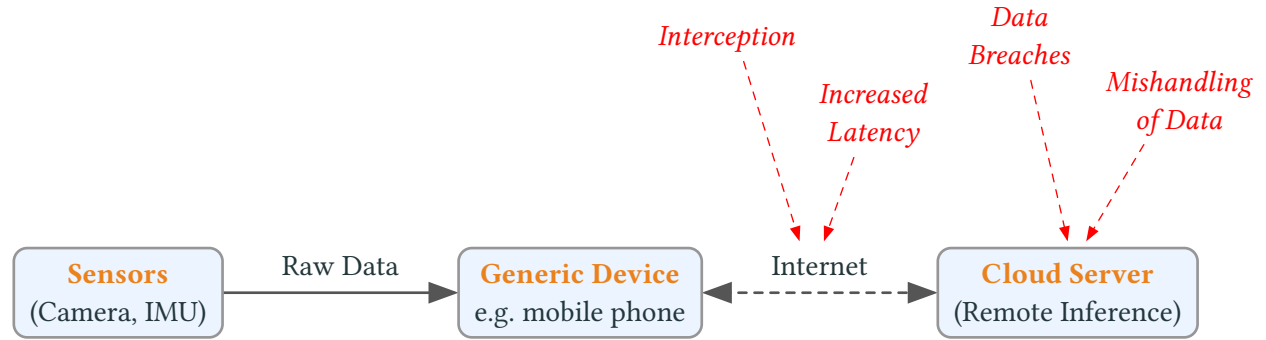
A Comparative Analysis of Memory Management, Concurrency, and Performance in Edge-AI

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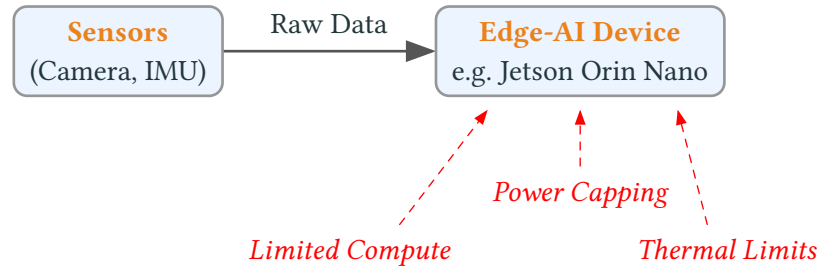
University of Dundee

Cloud vs Edge AI

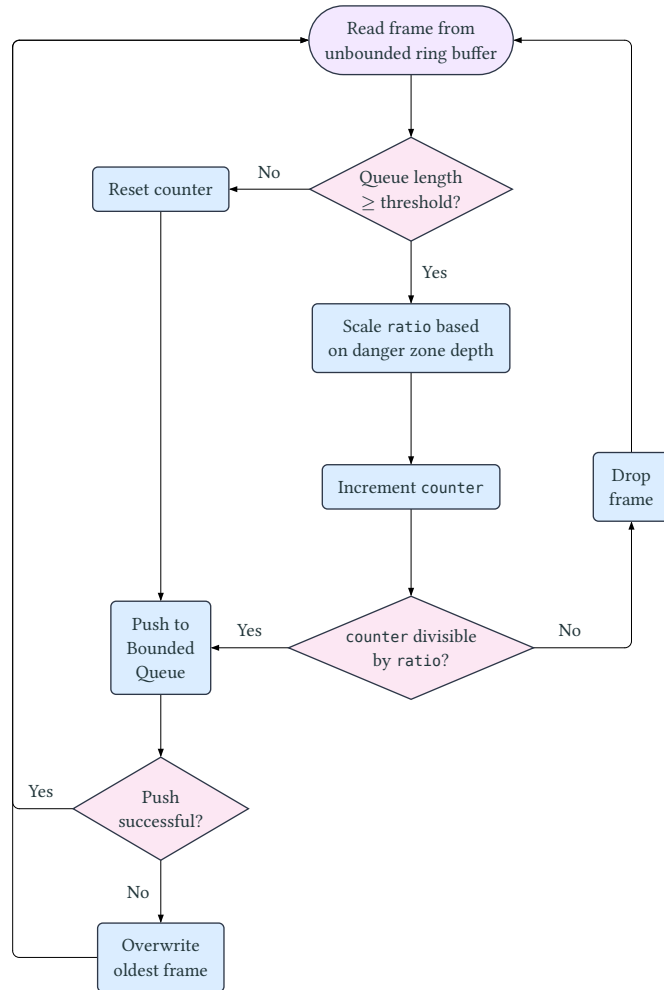
Cloud-AI Remote Inference



Edge-AI Local Inference



Backpressure Policies



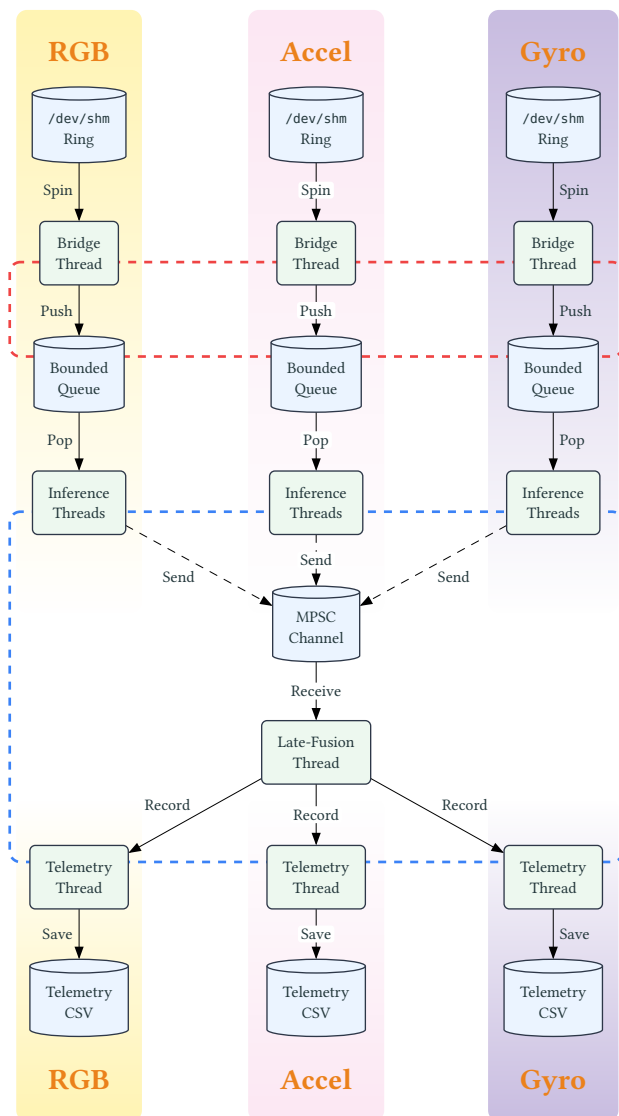
Flow Control:

- **Bounded Queue:** Blocks until space becomes available in the consumer buffer.
- **Exponential Backoff:** Sleeps for an exponentially increasing time period before retrying to push data into the consumer buffer.

Load Shedding:

- **Drop Oldest:** Drops the oldest data in the consumer buffer to make room for new data.
- **Drop Newest:** Drops incoming data when the buffer is full.
- **Adaptive Decimation:** Dynamically downsamples the data stream.

The Architecture



Data Streams (3): one exists for each sensor in use. Zero-allocation after initialisation.

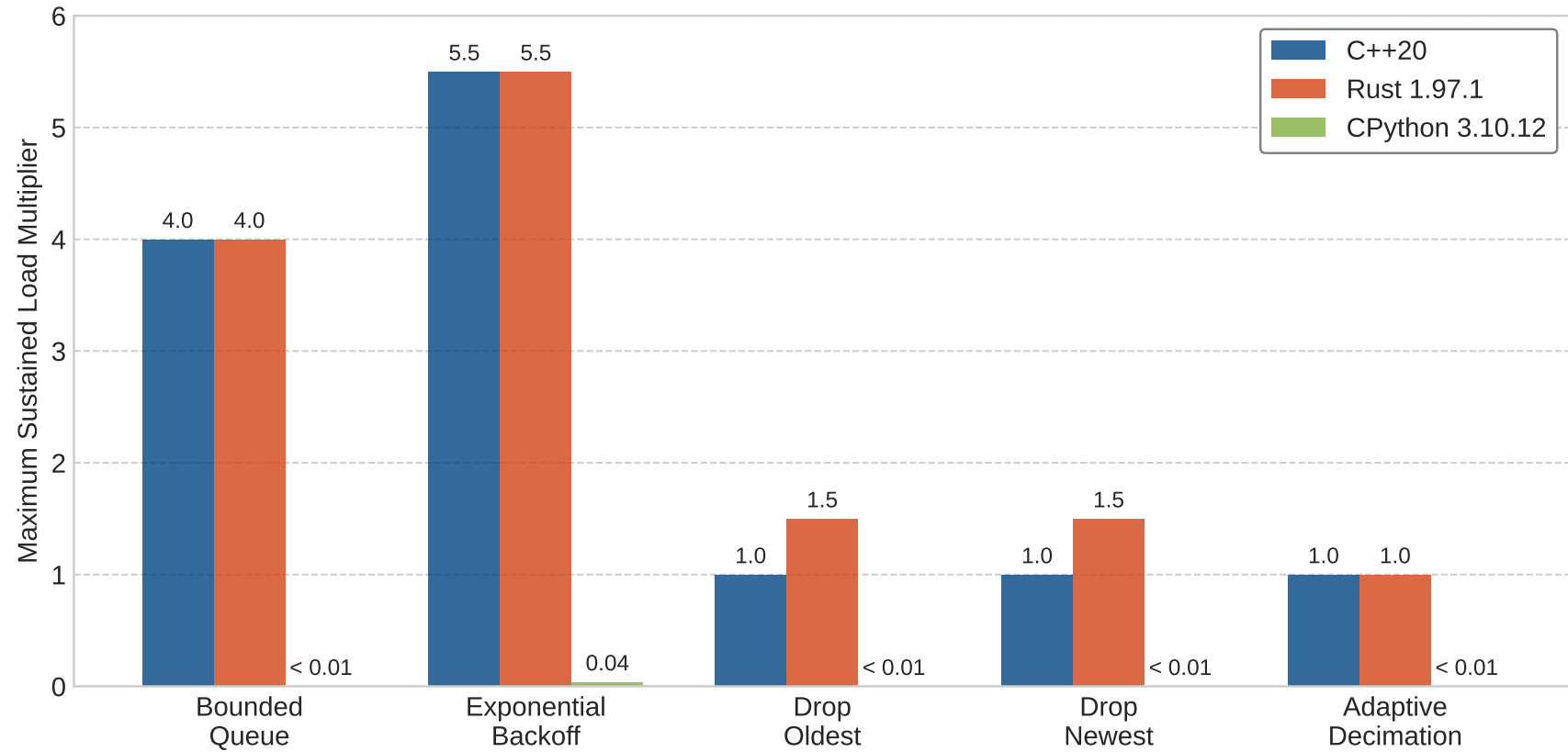
Bridge Threads (3): spin-waits on /dev/shm, attempts to add ingested frames into bounded buffer, applying backpressure policy when necessary.

Inference Threads (3): pulls frames from the bounded queue to create temporal windows and execute the ONNX model, pushing the result to the MPSC channel.

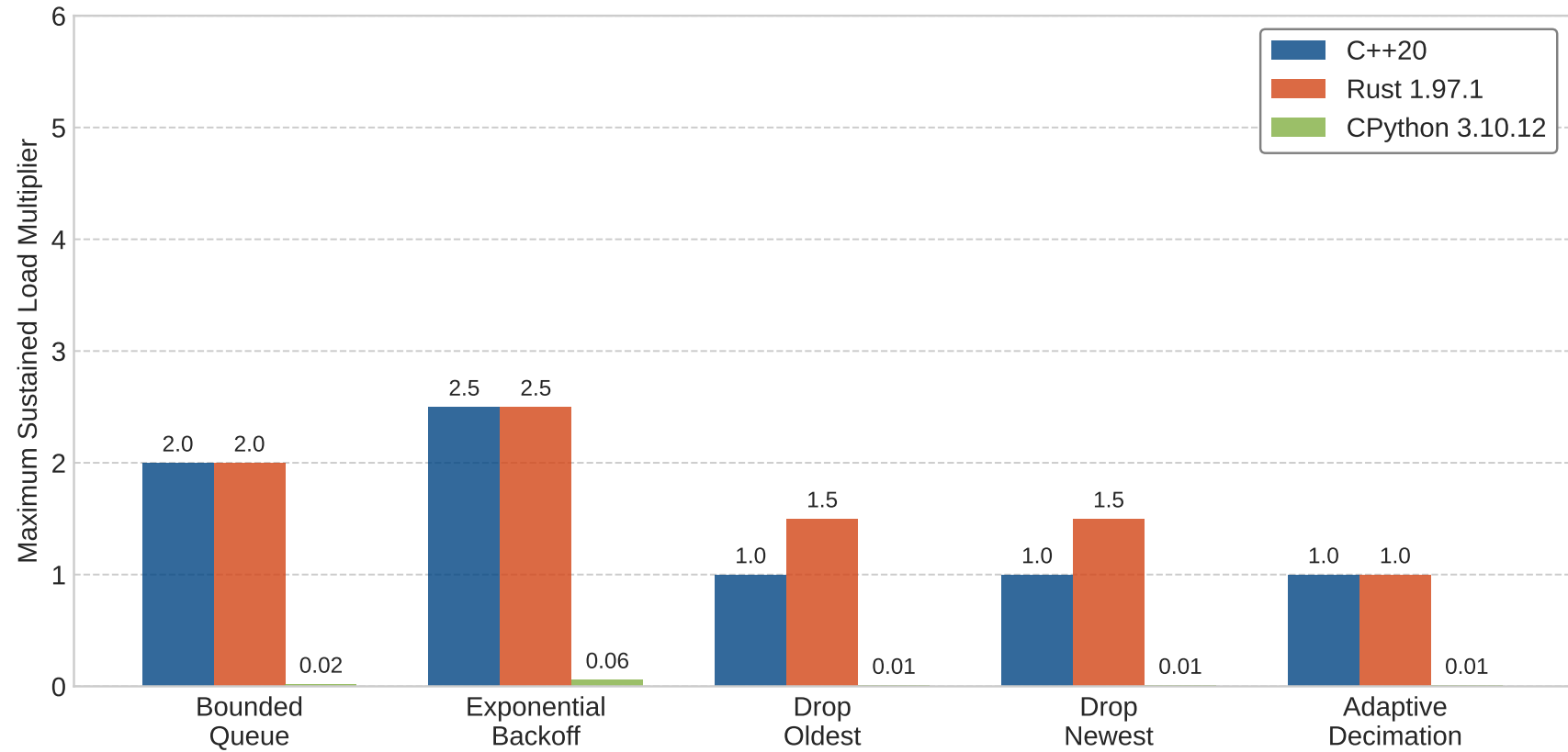
Late-Fusion Thread (1): consumes from the MPSC channel, anchored to the RGB stream. Pushes to the telemetry threads.

Telemetry Threads (3): uses HDR Histogram to calculate telemetry for later analysis, utilising a triple-buffered channel for zero-wait on the hot thread.

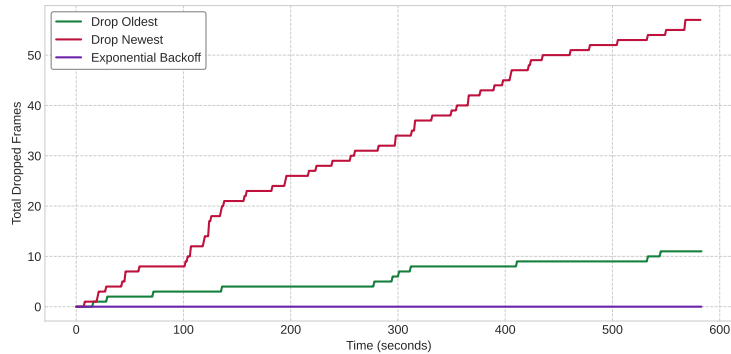
Baseline Performance (unconstrained MAXN_SUPER)



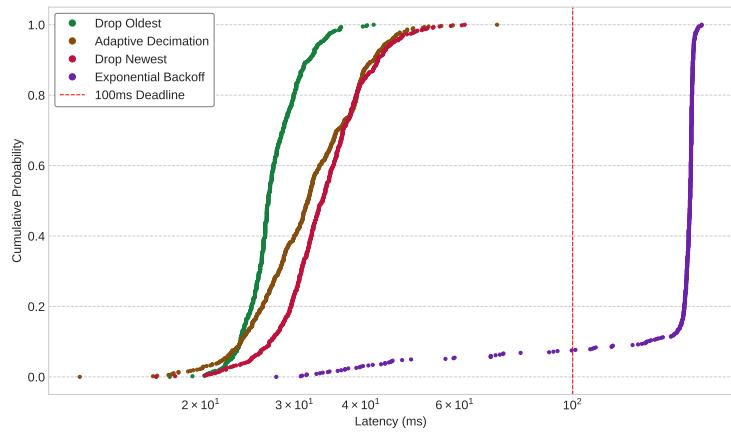
Baseline Performance (constrained 7-Watt)



Backpressure Trade-off



load = 2.5



load = 7.0

Moderate Load: Flow-control policies absorb the spikes by utilising the elasticity of the unbounded buffer, while load-shedding policies drop frames unnecessarily to maintain latency deadlines.

Severe Load: Using flow-control policies, the unbounded buffer fills to maximum capacity, pushing latency beyond the deadline.

Conclusion

